

Dirac-Monopole Dynamics on the Sphere: Magnetic Circles and the Complete Covariant-Laplacian Spectrum

Route-A source-local certificate HCS-C331

3 September 2026

Abstract

We identify a conserved Poincaré vector and every magnetic circle on the unit sphere, including its least period and all zero-energy and zero-charge faces. This is the *Poincaré vector and every magnetic circle* stage.

1 The classical flow

Let $S^2 \subset \mathbb{R}^3$ be the oriented unit sphere and let $Jv = x \times v$ on $T_x S^2$. Fix $q \in \mathbb{Z}$, put $b = q/2$, and consider

$$\nabla_t \dot{x} = bJ\dot{x}, \quad |\dot{x}|^2 = 2E. \quad (1)$$

Theorem 1 (All magnetic trajectories). *The vector*

$$K = x \times \dot{x} + bx$$

is constant and satisfies

$$K \cdot x = b, \quad |K|^2 = 2E + b^2.$$

If $E > 0$, the trajectory is the oriented small circle cut out by $K \cdot x = b$ and its least positive period is

$$T(E, q) = \frac{2\pi}{\sqrt{2E + q^2/4}}. \quad (2)$$

If $E = 0$, the trajectory is stationary. If $q = 0$ and $E > 0$, the circle is a great circle.

Proof. Skew-symmetry of J preserves $|\dot{x}|$. The ambient acceleration is $\ddot{x} = bx \times \dot{x} - 2Ex$. Consequently,

$$\dot{K} = x \times \ddot{x} + b\dot{x} = bx \times (x \times \dot{x}) + b\dot{x} = 0.$$

Orthogonality gives the two displayed identities. The triple-product identity also gives $K \times x = \dot{x}$, so x is a rigid rotation about the fixed axis K with angular speed $|K|$. Its squared circle radius is $2E/(2E + b^2)$, positive exactly when $E > 0$. A nonconstant circle first returns with its tangent after angle 2π , proving (2). The two boundary statements follow directly. $\square$

Time reversal gives the precise sign pairing: if $x_q(t)$ solves charge q , then $x_q(-t)$, with its initial velocity reversed, solves charge $-q$ and traverses the same geometric circle oppositely. This does not identify the solutions having the same initial velocity. The conserved-plane proof does not rely on coordinates or a gauge potential.

Source lineage

This paper is a source-local reconstruction and makes no priority claim.

References

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